Final Exam

● Graded

Student

Jingyang Zheng

Total Points

37.5 / 45 pts

Question 1

Overall Grades **37.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 7.5 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Jingyang Zheng	jz547	Jingyang Zheng

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Jingyang Zheng

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	3	4	5	5	0.5	5	37.5

Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

Write the system as a augmented matrix

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & -3 & -4 & -1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 5 \end{array}\right] \xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

Then $z=-5$, $y=7$, $x=11$, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1) $\text{col}(A) = \text{span}\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix} \right\}$ $\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_1 \leftarrow \frac{1}{2}R_1$, $R_2 \leftarrow -\frac{1}{5}R_2$ $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$R_2 \leftarrow R_2 - R_1$, $R_4 \leftarrow R_4 - R_1$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & -2 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ $R_2 \leftarrow R_2 + 2R_3$, $R_4 \leftarrow R_4 - R_3$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\text{rank}(A) = 2$, $\begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix}$ is redundant

therefore, a basis of $\text{col}(A)$ is $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) $\text{row}(A) = \text{span}\left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \right\}$

$\begin{bmatrix} 2 & -5 & 0 & 1 \\ 4 & 0 & 1 & 3 \\ 2 & 5 & 1 & 2 \end{bmatrix}$ $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$ $\begin{bmatrix} 2 & -5 & 0 & 1 \\ 0 & 10 & 1 & 1 \\ 0 & 10 & 1 & 1 \end{bmatrix}$ $R_3 \leftarrow R_3 - R_2$ $\begin{bmatrix} 2 & -5 & 0 & 1 \\ 0 & 10 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$ is redundant. Therefore, a basis of $\text{row}(A)$ is $\left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} \right\}$

(3) let $\left[\begin{array}{ccc|c} 2 & 4 & 2 & 0 \\ -5 & 0 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \end{array}\right]$, from (1) we know that $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

Assume that a solution for the system is $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $y + z = 0$, $x + 2y + z = 0$, let $z = t$

Therefore, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} t$

A basis of the null space of A is $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

5/5

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

Because $T: \mathbb{R}^3 \to \mathbb{R}^2$, A is a 2×3 matrix

Assume $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

Because $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} b-c \\ e-f \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} a-c \\ d-f \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -c \\ -f \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$$\begin{cases} b-c=1 \\ e-f=1 \\ a-c=0 \\ d-f=2 \\ -c=-f=2 \end{cases} \qquad \begin{cases} a=-2 \\ b=-1 \\ c=-2 \\ d=0 \\ e=-1 \\ f=-2 \end{cases}$$

$A = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$

3/5

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$, $\det(A-\lambda I) = 0$

$A-\lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$, $\det(A-\lambda I) = -(\lambda-1)^2(\lambda-2) = 0$

$\lambda_1 = 1$, algebraic multiplicity $= 2$; $\lambda_2 = 2$, algebraic multiplicity $= 1$

The Jordan canonical form of A is $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

why can't we have $\begin{bmatrix} 1 & & \\ & 1 & 2 \end{bmatrix}$??

-2

4/5

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

The quadratic form of $f(x,y,z) = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} = A$

Find the eigenvalues and eigenvectors of A

$A-\lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$, $\det(A-\lambda I) = -(\lambda-3)^2(\lambda-4) - 2(3-\lambda) = -(\lambda-2)(\lambda-3)(\lambda-5) = 0$

$\lambda_1 = 2$, $\lambda_2 = 3$, $\lambda_3 = 5$

① $A-2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, let $\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & -1 & 2 & 0 \end{array}\right]$ $R_3 \leftarrow R_3 - R_1$ $\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right]$

$R_3 \leftarrow R_3 + R_2$ $\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$, let $\vec{v_1} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $\begin{cases} x_1 + z_1 = 0 \\ y_1 - z_1 = 0 \end{cases}$, $\vec{v_1} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

② $A-3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$, let $\left[\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \end{array}\right]$ $R_2 \leftarrow -R_2$, $R_1 \leftarrow R_3$ $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$

$R_2 \leftarrow R_3 - R_2$ $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$, let $\vec{v_2} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$, $\begin{cases} z_2 = 0 \\ x_2 - y_2 + z_2 = 0 \end{cases}$, $\vec{v_2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

③ $A-5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$, let $\left[\begin{array}{ccc|c} -2 & 0 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 1 & -1 & -1 & 0 \end{array}\right]$ $R_1 \leftarrow R_1 + 2R_3$ $\left[\begin{array}{ccc|c} 0 & -2 & -1 & 0 \\ 0 & -2 & -1 & 0 \\ 1 & -1 & -1 & 0 \end{array}\right]$

$R_1 \leftrightarrow R_3$ $\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & -2 & -1 & 0 \end{array}\right]$ $R_3 \leftarrow R_3 - R_2$ $\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$, let $\vec{v_3} = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$, $\begin{cases} x_3 - y_3 - z_3 = 0 \\ -2y_3 - z_3 = 0 \end{cases}$, $\vec{v_3} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

let $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ t \end{bmatrix}$, $x = -u+v+t$, $y = u+v-t$, $z = u+2t$

$f(x,y,z) = f(u,v,t) = 6u^2 + 6v^2 + 30t^2$

Therefore, after diagonalizing, the quadratic form becomes $f(x,y,z) = X^2 + Y^2 + 5Z^2$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

Because $\vec{w} = a\vec{v_1} + b\vec{v_2} + c\vec{v_3}$, $\begin{cases} 2b - c = -3 \\ 2a - b + c = 8 \\ -2a + 2b = -8 \end{cases}$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 + R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_2 - 2R_3} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & -3 & -3 \end{array}\right], \quad c = 1, \quad b = -1, \quad a = 3$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Using the Gram-Schmidt orthogonalization,

$\vec{u_1} = \vec{v_1} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, $\vec{u_2} = \vec{v_2} - \frac{\vec{u_1}\cdot\vec{v_2}}{\|\vec{u_1}\|^2}\cdot\vec{u_1} = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{1^2+1^2}\cdot\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

let $\frac{\vec{u_1}}{\|\vec{u_1}\|} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} \times \frac{1}{\sqrt{1^2+1^2}} = \frac{\sqrt{2}}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, $\frac{\vec{u_2}}{\|\vec{u_2}\|} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix} \times \frac{1}{\sqrt{3^2+(-1)^2+1^2+(-1)^2}} = \frac{\sqrt{3}}{6}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

An orthonormal basis of V is $\left\{\frac{\sqrt{2}}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \frac{\sqrt{3}}{6}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}\right\}$

0.5/5

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

Because $A^T = -A$, therefore $\begin{cases} \text{when } i \neq j,\ a_{ij} = -a_{ji} \\ \text{when } i = j,\ a_{ij} = 0 \end{cases}$ $(i \leq n,\ j \leq n,\ i \in N^*, j \in N^*)$

If $n=1$, $A = [0]$, $\det A = 0$

If $n > 1$, $A = \begin{bmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ -a_{12} & 0 & a_{23} & \cdots & a_{2n} \\ -a_{23} & -a_{13} & 0 & \cdots & a_{3n} \\ \vdots & & & & \\ -a_{1n} & -a_{2n} & -a_{3n} & \cdots & 0 \end{bmatrix}$

let $R_2 = a_{23} R_2 = [-a_{12}a_{23} \quad 0 \quad a_{23}^2 \cdots \quad a_{2n}a_{23}]$

$R_3 = a_{12} R_3 = [-a_{12}a_{23} \quad -a_{12}a_{13} \quad 0 \cdots \quad a_{12}a_{3n}]$

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

Because $\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$, $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$

Therefore, $\|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\vec{u}\cdot\vec{v} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\vec{u}\cdot\vec{v}$

$\vec{u}\cdot\vec{v} = 0$, $\vec{u}$ and $\vec{v}$ are perpendicular